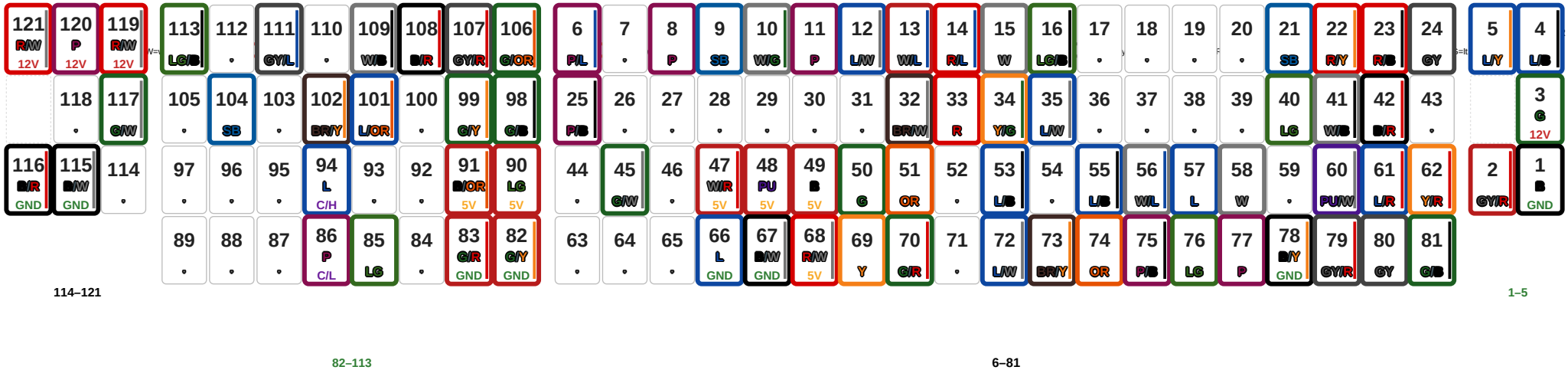
2005 Nissan 350Z VQ35DE — Engine harness print pack

Motor-control loom on the bench (F-series) · ECM MEC61-510 · 2 pages: EC-123 + cards

Card · color = outline · 5V amber / 12V red / GND green · ECM Inverted L/R · cards FSM T.S. · FSM EC 2005 · OEM · not a service manual

1 · EC-123 INVERTED — MEC61-510 · (same sheet as cover)

pin # top · wire code bottom · border = wire · red = bench · Inverted L/R



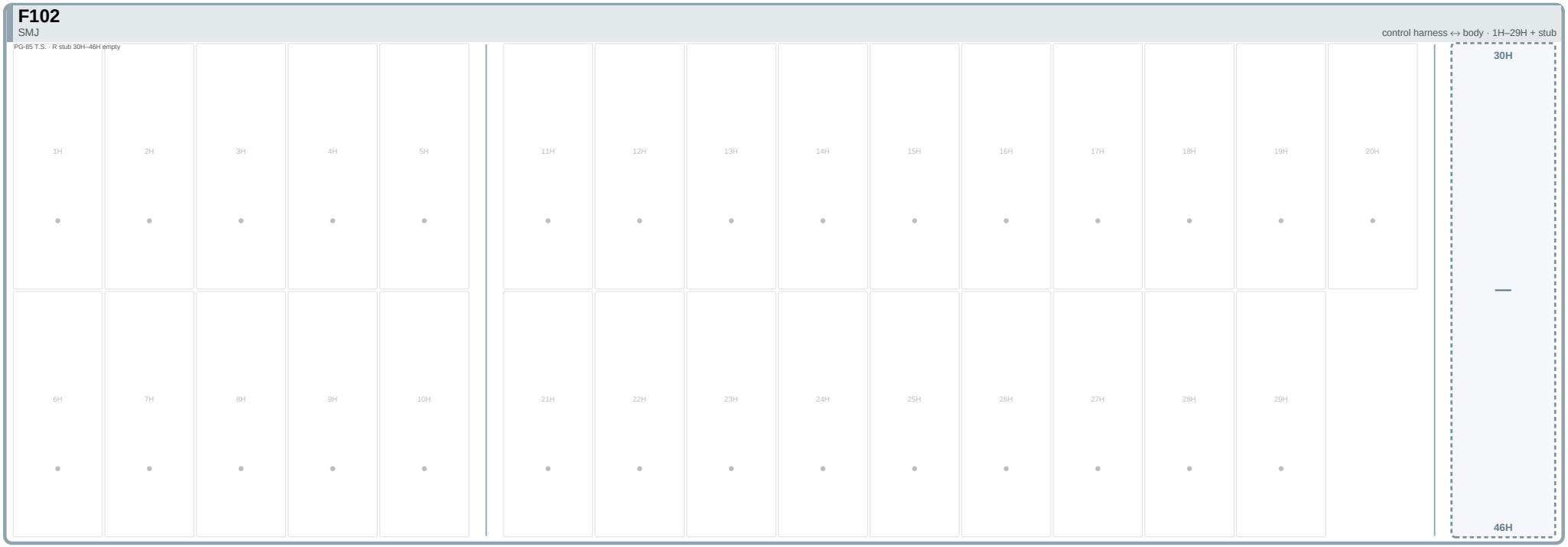
ECM RAILS · lines leaving the module (engine harness)

pin border = wire color · FSM EC-123

12V	BATT 121=R/W (E108·68G → F102·5H) · IGN 119=R/W · IGN 120=P · VMOT 3 (E8·42)
5V	47 TPS/ETC · 48 EVAP · 49 A/C press · 68 PSP · 90 APP1 · 91 APP2 (separate rails, not one shared 5V)
GND	ECM 1=B · 115=B/W · 116=B/R (→F103·3/2/4→F152) · SNS 66/67/78/82/83 · signals: KNK15 · CKP13 · CAN 86/94 · AFh 2/24

F102 · SMJ (ENGINE CONTROL HARNESS · PG-85) · not inverted · motor face of harness

PG-85 T.S. · 0 cavities · A/T = automatic only: 28H PNP · 23H START (EC-644) · 24H K-TCM (AT-194); empty on M/T (EC-646)



2 · Motor cards · sensor → intermediate F → ECM · group outline · 5V amber · FSM T.S. faces

cavity: id · color · →ECM · 5V=amber ring · 12V red / GND green

PATH A/F + HO2S · sensors → E12■F3 (heaters / power) → ECM · heaters ECM 2 / 24

A/F B1
sensor — FSM T.S. · pass/RH

→35	fuse	→16
L/W	12V	LG/B
5	3	1
→56	→2	→75
W/L	GY/R	P/B
6	4	2

A/F B2
sensor — FSM T.S. · driver/LH

→57	fuse	→76
L	12V	LG
5	3	1
→58	→24	→77
W	GY	P
6	4	2

F11
HO2S B1 — post-cat RH

→25	fuse
P/B	12V
htr	12V

F12
HO2S B2 — post-cat LH

→6	fuse
P/L	12V
htr	12V

A/F heaters / coil·inj 12V rail: E12■F3 (mate in joints below)

PATH Knock · sensor → F14/F229 → ECM 15 · shield → B/R 116 → F103·4 → F152 · other motor sensors (direct → ECM)

Knock
sensor

→116	→15
B	W
GND	15S

F14
knock

→116	→15
B	W
GND	15S

F229
knock

→15	→116
W	B
15S	GND

F25
MAF

pwr	→67	→51
R/W	B/W	OR
12V	GND	51S

F10
CKP

gnd	→13	12V
B	W/L	R/W
GND	13S	PWR

CMP B1
cam

12V	→33	gnd
R/W	R	B
12V	33S	GND

CMP B2
cam

12V	→14	gnd
R/W	R/L	B
12V	14S	GND

ECT
coolant

→67	→73
B/W	BR/Y
GND	73S

IAT
inside MAF F25

→34	→67
Y/G	B/W
IAT	GND

ETC
throttle — F31 · 1=5V 2=T2 3=M- 4=T1 5=GND 6=M+

→4	→69	→47
L/B	Y	W/R
3	2	1
→5	→66	→50
L/Y	L	G
6	5	4

VTC B1
intake — intake cam RH · →ECM 11

ign	→11
W/L	W/R
12V	ECM

VTC B2
intake — intake cam LH · →ECM 10

ign	→10
W/L	W/G
12V	ECM

Coil 1
→ECM 62

ign	F23	→62
W/L	B	Y/R
12V	GND	62S

Coil 2
→ECM 81

ign	F23	→81
W/L	B	G/B
12V	GND	81S

Coil 3
→ECM 61

ign	F23	→61
W/L	B	L/R
12V	GND	61S

Coil 4
→ECM 80

ign	F23	→80
W/L	B	GY
12V	GND	80S

Coil 5
→ECM 60

ign	F23	→60
W/L	B	PU/W
12V	GND	60S

Coil 6
→ECM 79

ign	F23	→79
W/L	B	GY/R
12V	GND	79S

Inj 1
→ECM 23

→23	ign
R/B	R
ECM	12V

Inj 2
→ECM 42

→42	ign
R/W	R
ECM	12V

Inj 3
→ECM 22

→22	ign
R/Y	R
ECM	12V

Inj 4
→ECM 41

→41	ign
R/L	R
ECM	12V

Inj 5
→ECM 21

→21	ign
L/W	R
ECM	12V

Inj 6
→ECM 40

→40	ign
PU/R	R
ECM	12V

Motor joints · F103/gnd4 → F152 (·1→F3·6→E17) · Ign condenser · SNS GND · E12■F3 · E10■F1 (battery tray)

F103
grounds — 2/3/4 → F152 · 1 → F3·6

F3·6	→115	→1	→116
B	B/W	B	B/R
1	2	3	4

F152
engine — body ground — ECM 1/115/116

ring B gnd

E17
body — body ground — via F103·1 → F3·6

ring B gnd

F16
condenser — noise · near cyl3

cyl3
—
~2µF

SNS
grounds — sensor grounds

→66	→67	→78
B	B	B
66	67	78

E12
power — 8-pin GY · VB 119/120 · IGN

F3
power — 8-pin GY · VB 119/120 · IGN

E10
tray — 9-pin GY · START · REV Y/R

F1
tray — 9-pin GY · START · REV Y/R